KUDZAI RUE

Question 1

```
SELECT DISTINCT dept
FROM students;
```

department
IT
HR
Finance

Question 2.

```
SELECT department,
    Avg (age) AS avg-age
FROM students
GROUP BY department;
```

Department	age
IT	20,5
HR	22
finance	23

Question 3

```
SELECT department,
    COUNT (*) AS student_count
    FROM students
    GROUP BY department
    HAVING COUNT (*) > 1;
```

department	student_count
IT	2
HR	2.

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Question 4.

```
SELECT student_id,
       name,
       age,
       department
WHERE age IS BETWEEN 21 AND 23;
```

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

Question 5

```
SELECT *
FROM students
WHERE age > 21 AND dept = 'IT' OR 'HR';
```

student_id	name	age	department
5	Eve	22	HR.

Question 6

```
SELECT department,
    SUM (credits) AS total_credits
FROM students
GROUP BY department;
```

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Question 6

```
SELECT department,
       SUM (credits) AS total_credits
FROM ~~students~~ courses
WHERE total credits ≥ 5
GROUP By department;
```

department	total_credits
IT	11

~~Question 7~~

```
~~SELECT *~~
~~FROM courses~~
~~WHERE credits > 4~~
~~WHERE credit < 4~~
```

Question 7

```
SELECT *
FROM courses
WHERE credit NOT > 4
```

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	finance	2
105	statistics	HR	3.

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Question 8

```
SELECT courseid,
       course_name,
       credits
FROM courses
ORDER BY credits desc
LIMIT 3;
```

course_id	course_name	credits
102	Python	4
103	Data science	4
105	Statistics	3

Question 9

```
SELECT MAX (grade) AS max_grade,
       MIN (grade) AS min_grade,
       AVG (grade) AS avg_grade
FROM enrollments;
```

Max_grade	min_grade	avg_grade
90	78	84,6

Question 10

```
SELECT course_id,
       Count (*) AS enrollment_count
FROM enrollments
Group By course_id;
```

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course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

Question 11

```
SELECT department,
    SUM(salary) AS total_salary,
    SUM(Bonus) AS total_bonus,
    FROM salaries
    GROUP BY department;
```

department	total_salary	total_bonus
IT	102000	10500
HR	109000	7500
FINANCE	70000	6000

Question 12.

```
SELECT department,
    AVG(salary) AS AVG_salary
FROM salaries
WHERE AVG_salary > 55000
GROUP BY department;
```

Department	AVG_salary
IT	61000
finance.	70000.

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Question 13

```
SELECT employee_id,
       name,
       salary,
       bonus,
  SUM (salary, Bonus)  AS  total_compensation
FROM salaries
WHERE  SUM (salary, Bonus) > 60000;
```

employee_id	name	salary	bonus	total-compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

Question 14

```
SELECT department,
       SUM (Budget)  AS  total_budget,
       AVG (budget)  AS  ~~Avg~~ AVG-budget
FROM  Project
HAVING AVG (budget) > 70 000;
```

department	total_budget	AVG-budget
IT	270000	135 000
finance	80000	80000

Question 15

```
SELECT projectid,
       project_name,
       department,
       budget
FROM projects.
```

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WHERE budget BETWEEN 50000 AND 120000 AND
department NOT marketing;

Project id	Project_name	department	budget
1	AI App	IT	120000
2	Payroll system	finance	80000
5	HR Portal	HR	50000.

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